Becoming an IR Rock Star

(Superstars 2.0)

Shawnee State University

Matt Scheurer

http://slides.dfirmatt.com

```
if (badEvent == "true") {
  perform = "Incident Response";
}
```

About Me

I work for a big well-known organization...

As Vice President (VP) of Computer Security and Incident Response (IR). However, I have many years of hands-on technical experience, including Digital Forensics & Incident Response (DFIR).

https://www.linkedin.com/in/mattscheurer

https://x.com/c3rkah

Where I volunteer...

I am an Official

HACKING IS NOT A CRIME

Advocate
https://www.hackingisnotacrime.org

WARREN COUNTY CAREER CENTER

Advisory Board: Information Technology and Cybersecurity
https://www.mywccc.org/

WOMSA
Empowering Women in Cybersecurity

Women's Security Alliance (WomSA) Technical Mentor
https://www.womsa.org

Disclaimer!

Yes, I have a day job. However…

Opinions expressed are based solely on my own independent security research and do not express or reflect the views or opinions of my employer.

Agenda

1) Incident Response (IR) Essentials

2) Q&A

Slide Deck

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Agenda

1) Incident Response (IR) Essentials

2) Q&A

Why this talk?

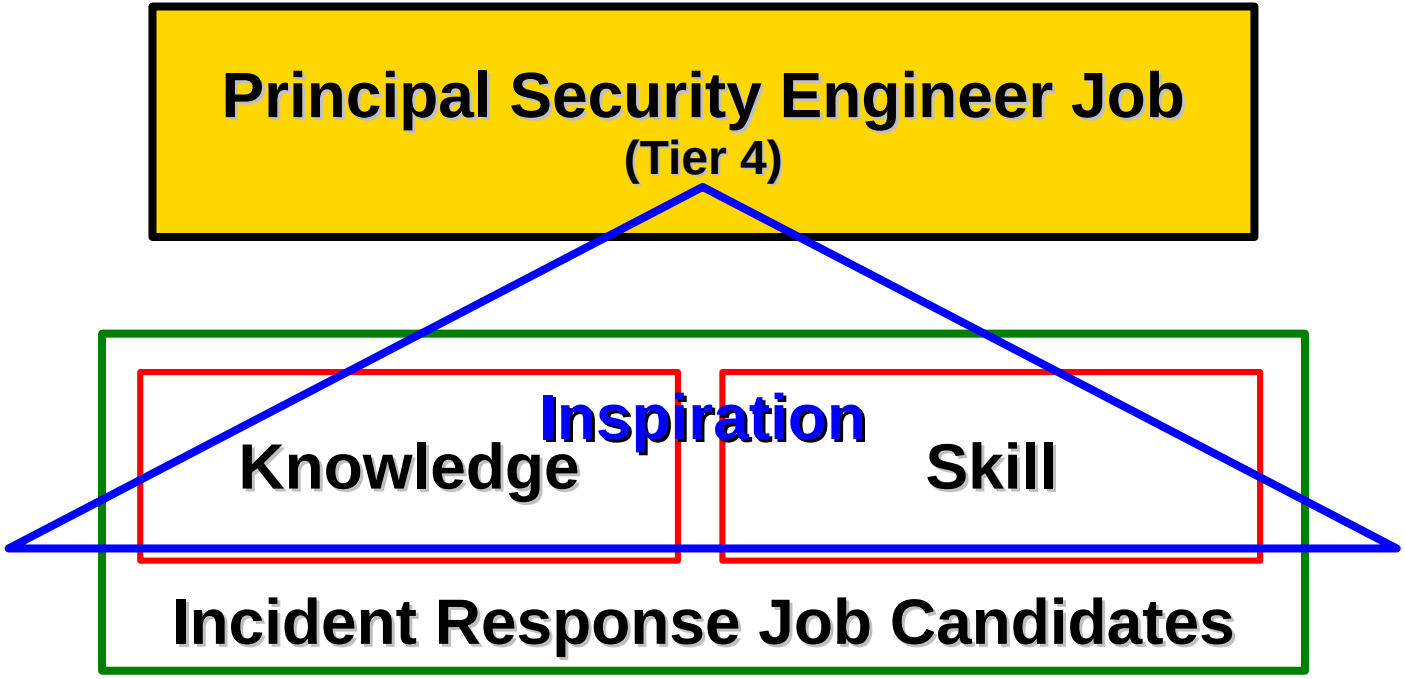
Principal Security Engineer Job
(Tier 4)

Knowledge

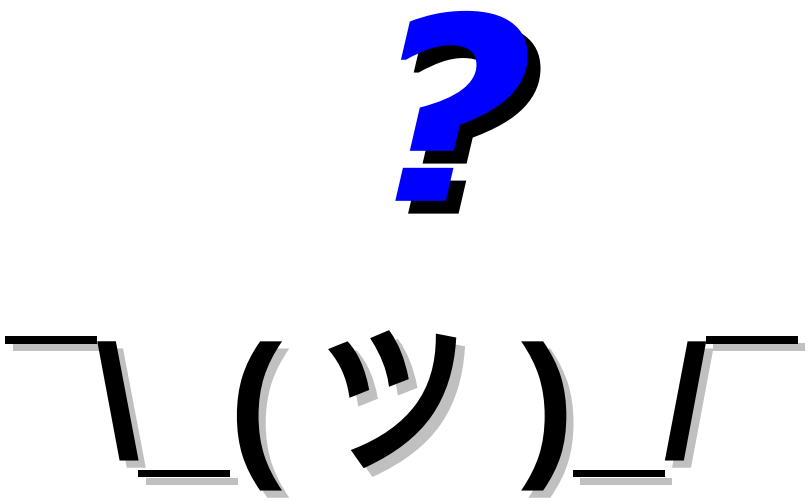
Skill

Incident Response Job Candidates

Why this talk?



Role of Incident Response

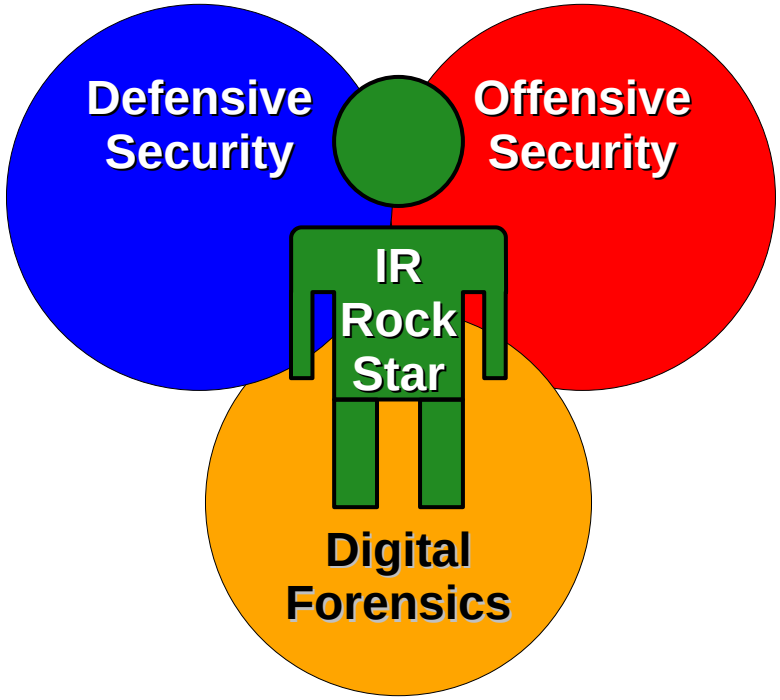


Security Investigation

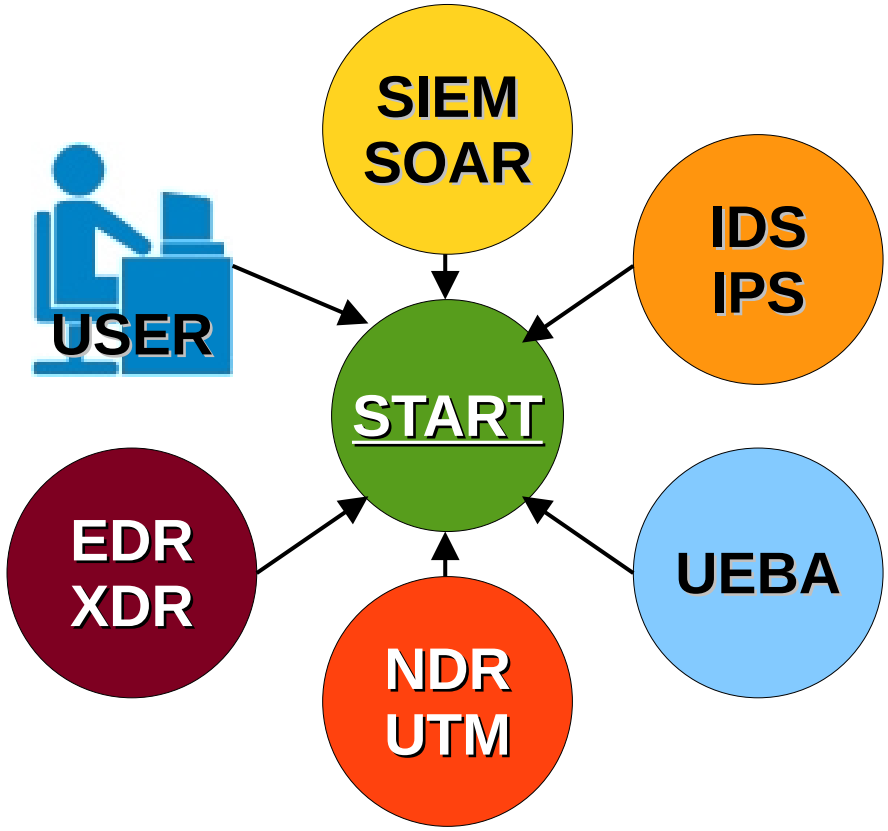
- Who
- What
- When
- Where
- Why
- How

Aspiring Incident Responders

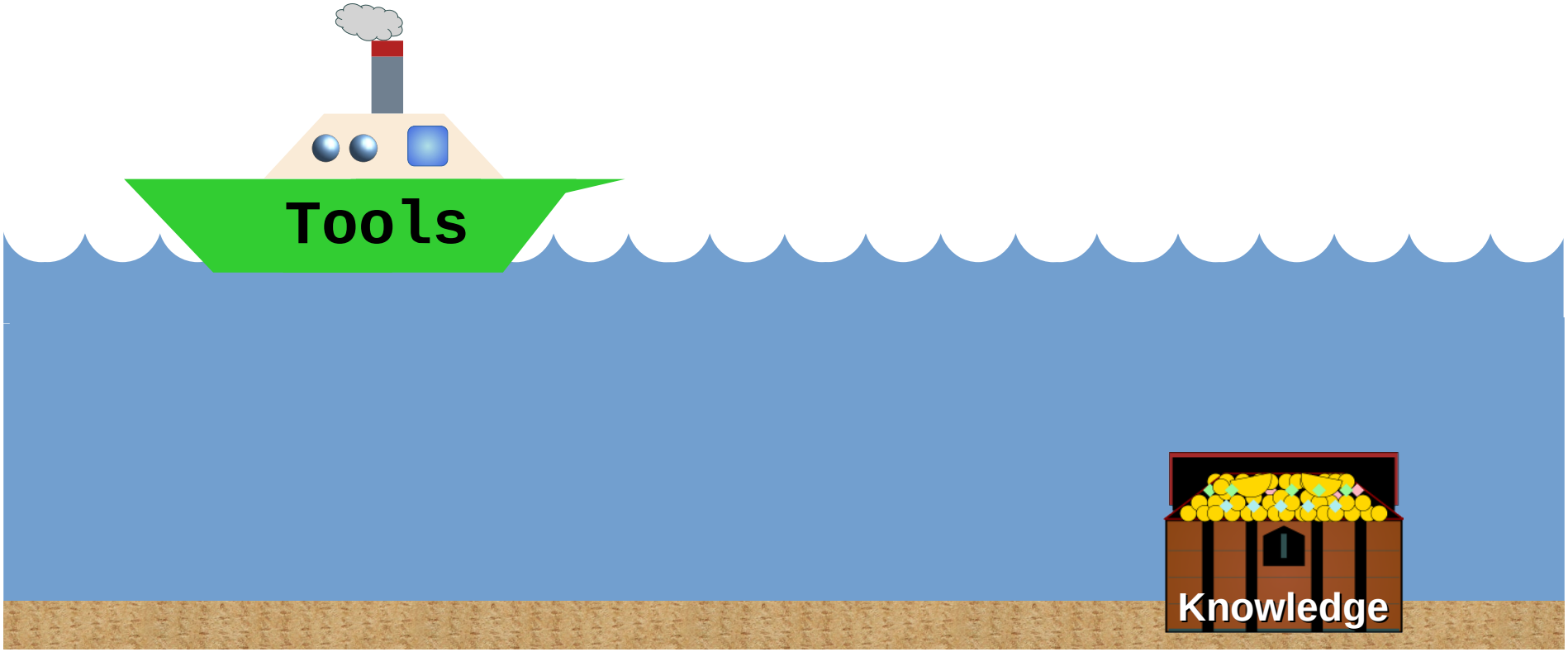
Incident Responders



When IR starts



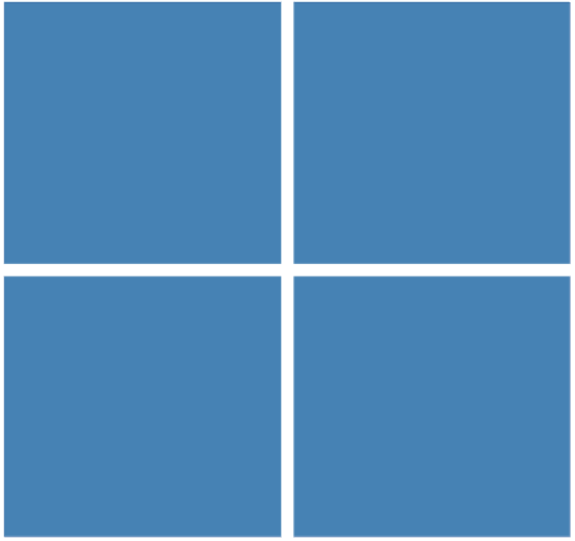
Diving Deep

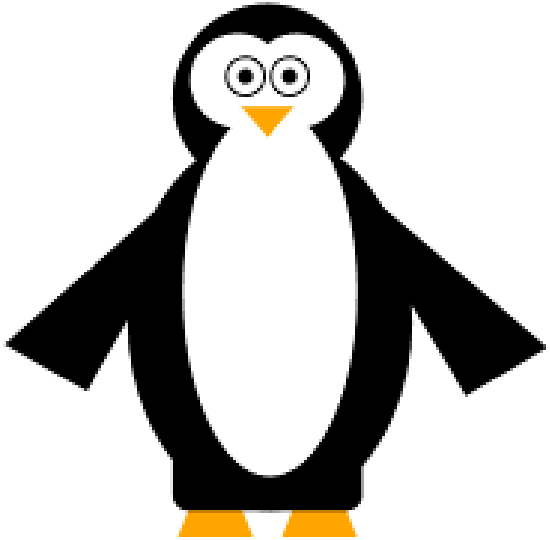


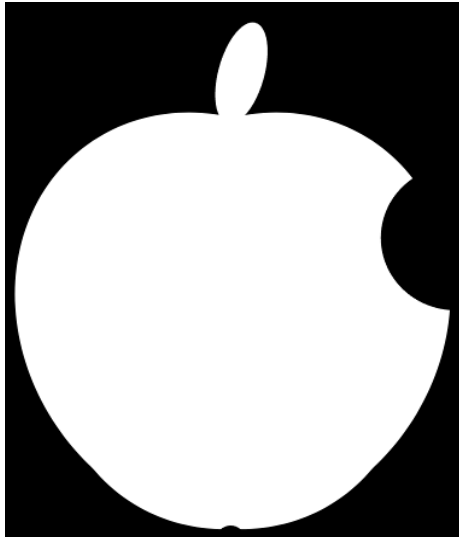
Alert Dispositions

	True (Detection)	False (Detection)
Positive (Condition)	**True Positive:** Correctly classifying malicious activity as malicious	**False Positive:** Incorrectly classifying benign activity as malicious
Negative (Condition)	**True Negative:** Correctly classifying benign activity as benign	**False Negative:** Incorrectly classifying malicious activity as benign

Operating Systems





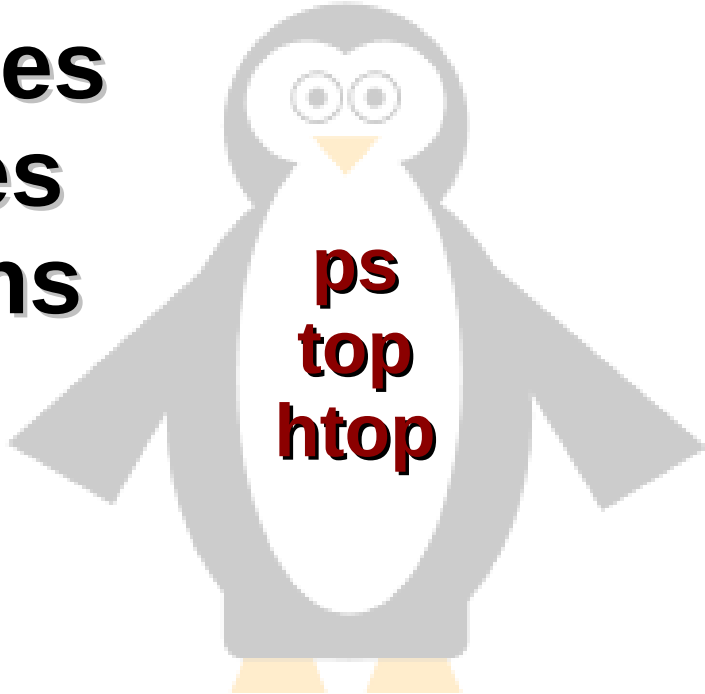


OS Processes & More

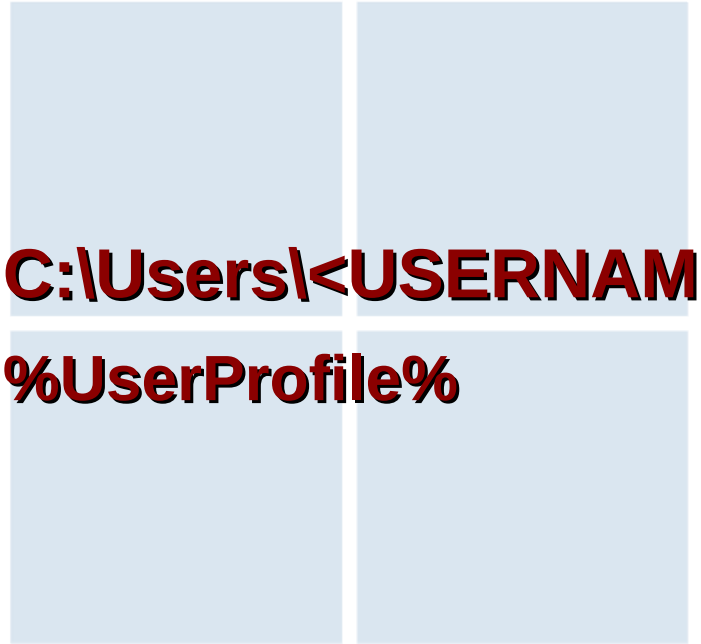
Task Manager
(taskmgr.exe)

Services
(services.msc)

Processes
Services
Daemons



Default User directory



C:\Users\<USERNAME>

%UserProfile%

/home/<USERNAME>

Windows Filesystem

Configuration

Registry (regedit.exe)

Hive files

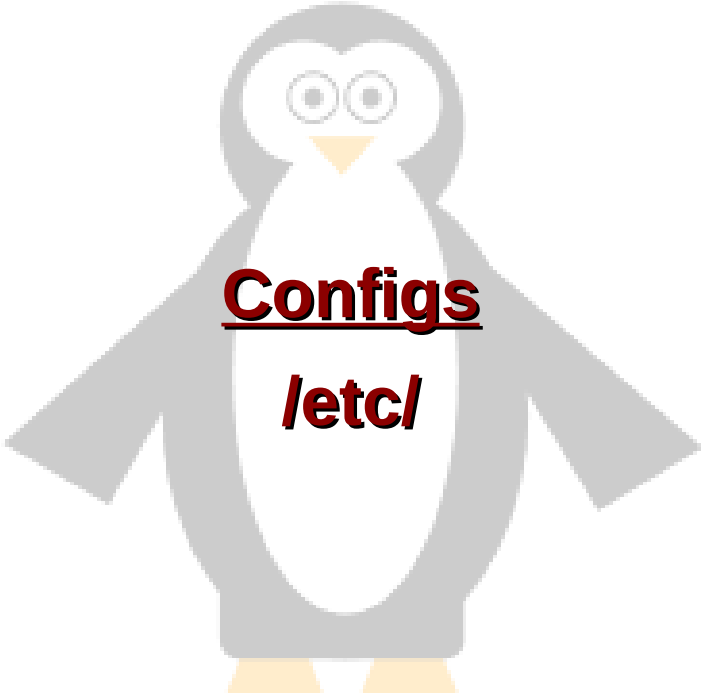
C:\Windows\System32\

- \config
- \config\RegBack

%UserProfile%

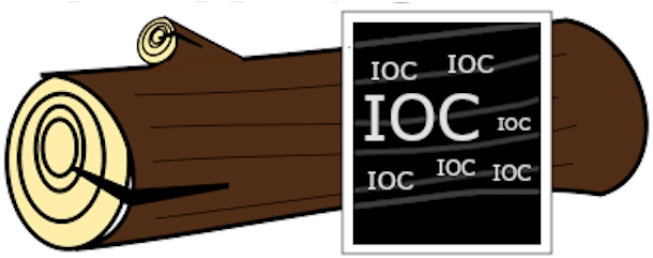
- \NTUSER.DAT

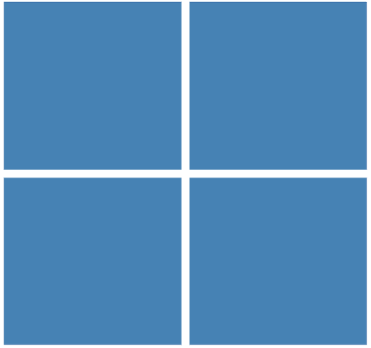
Linux Filesystem



- Linux Foundation
  - Filesystem Hierarchy Standard (FHS)

Various Log files





Windows Logs

Application

Security

Setup

System

etc.

C:\System32\winevt\Logs

- %System32%
- ***.evtx**

Event Viewer	**(eventvwr.exe)**
Event ID	**Logon type**

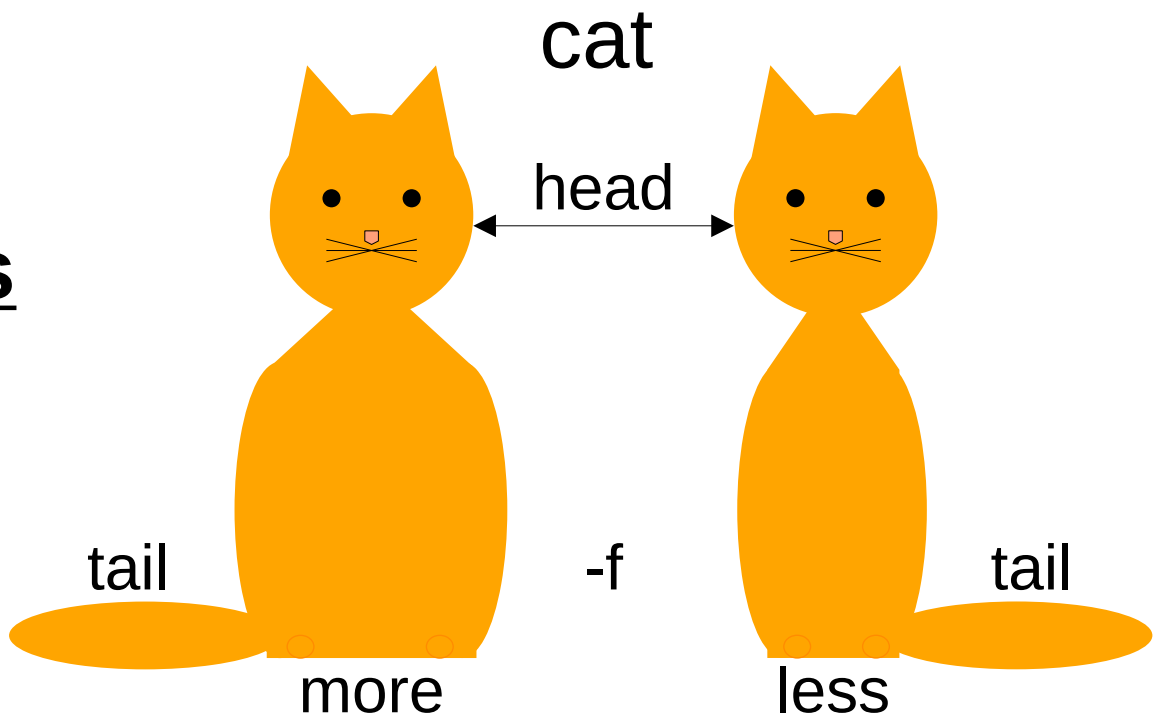
Linux Logs

Path

/var/log

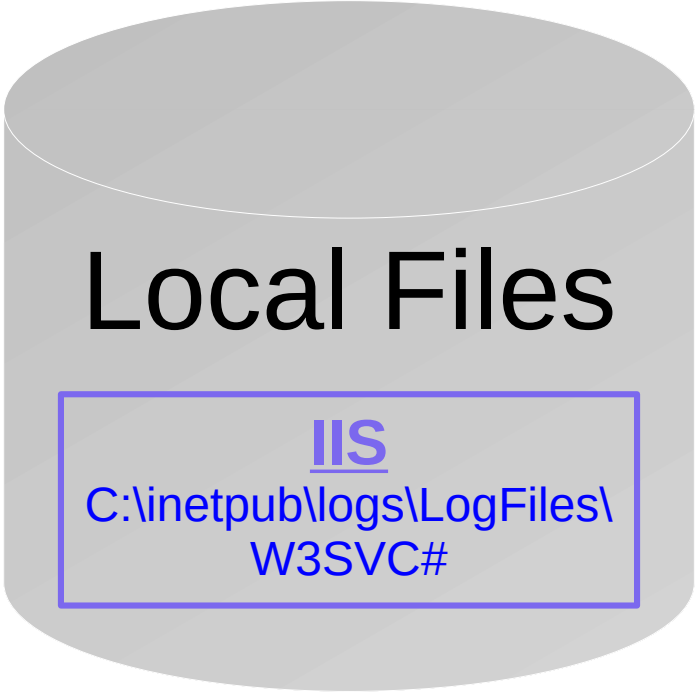
Epoch timestamps

01-01-1970



Server Logs

Host
Native
Logs

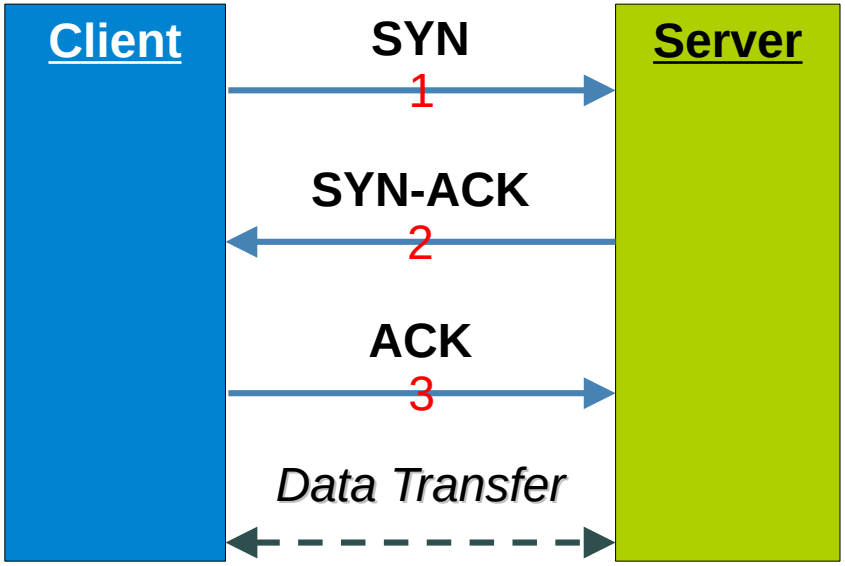

Local Files
IIS
C:\inetpub\logs\LogFiles\
W3SVC#

Application Logs

Centralized
Console

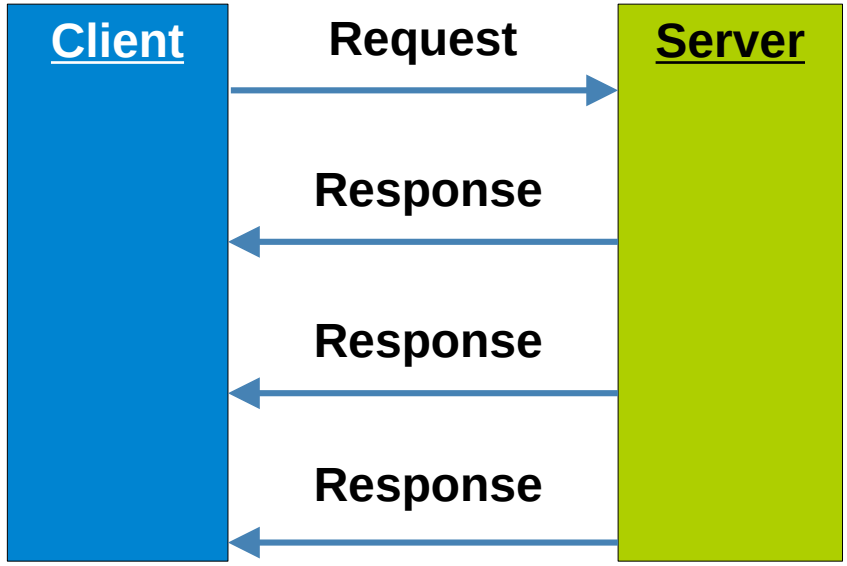
Network Transport Protocols

TCP



(3-way handshake)

UDP



(connectionless)

Computer Network Ports

TCP

22 (SSH)

80 (HTTP)

443 (HTTPS)

445 (SMB)

3389 (RDP)

UDP

53 (DNS)

67/68 (DHCP)

123 (NTP)

161 (SNMP)

Others

HTTP Request Methods

GET

HEAD

CONNECT

OPTIONS

POST

PUT

DELETE

TRACE

PATCH

HTTP Response Codes

200
series
(“Ok” codes)

300
series
(redirects)

400
series
(client errors)

500
series
(server errors)

OWASP Top 10

OWASP®

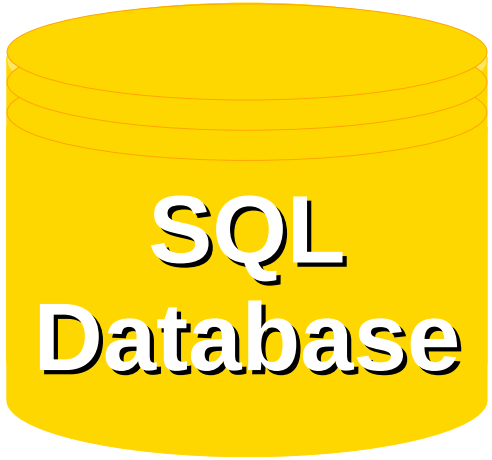
TOP 10

Open Web Application Security Project, Top 10

Structured Query Language

Free online training

- https://www.sqlcourse.com/beginner-course/
- https://www.sqlcourse.com/advanced-course/



Data Encoding

String	Hello attendees!
Base64	SGVsbG8gYXR0ZW5kZWVzIQ==
ROT13	Uryyb nggraqrrf!
HEX (from ASCII)	48 65 6c 6c 6f 20 61 74 74 65 6e 64 65 65 73 21

NOTE:	All 3 encoding methods are commonplace

IOCs & TTPs

Indicators of Compromise

- **Filenames & Paths**
- **Hashes**
- **IP addresses**
- **Domains & URLs**

File Execution?

Tactics, Techniques, and Procedures

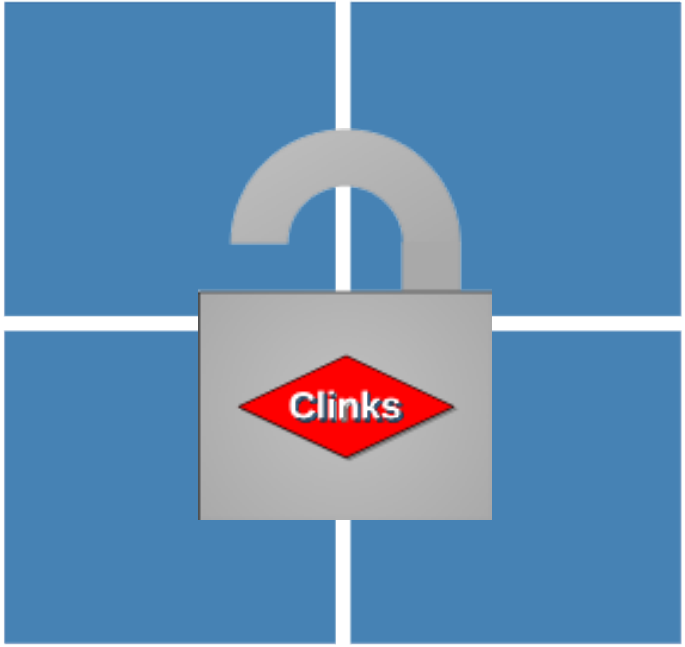
- **MITRE ATT&CK** framework
  - https://attack.mitre.org/

Persistence
TA0003

Windows Design & Security

50+ auto-executable file extensions

Ready!
Fire!
Aim?
...Whoops!!!

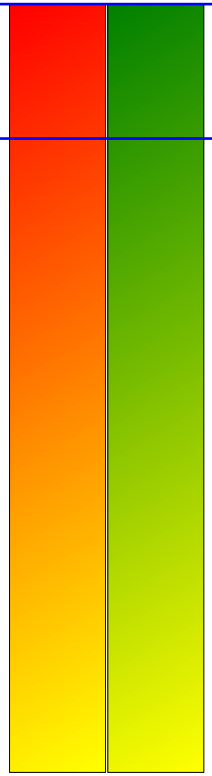


Volatile Data Collection

Most Volatile — Highest Priority

Order of Volatility

FIFO = First In, First Out



Least Volatile — Lowest Priority

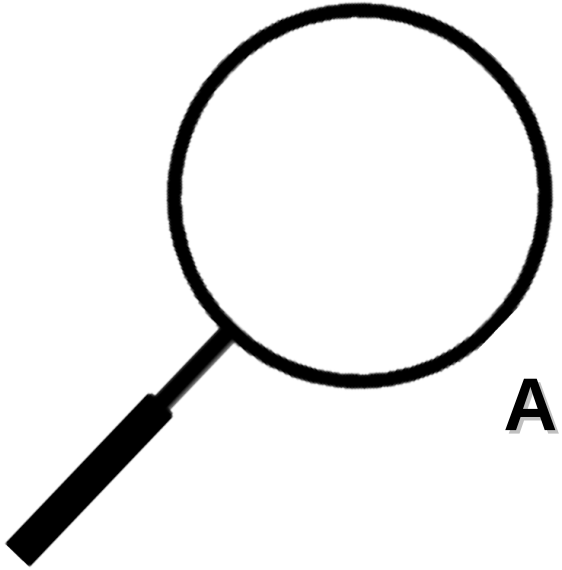
Industry Standards

RFC 3227

NIST SP 800-86

Digital Forensics

https://youtu.be/t4Vy3irI0Tk



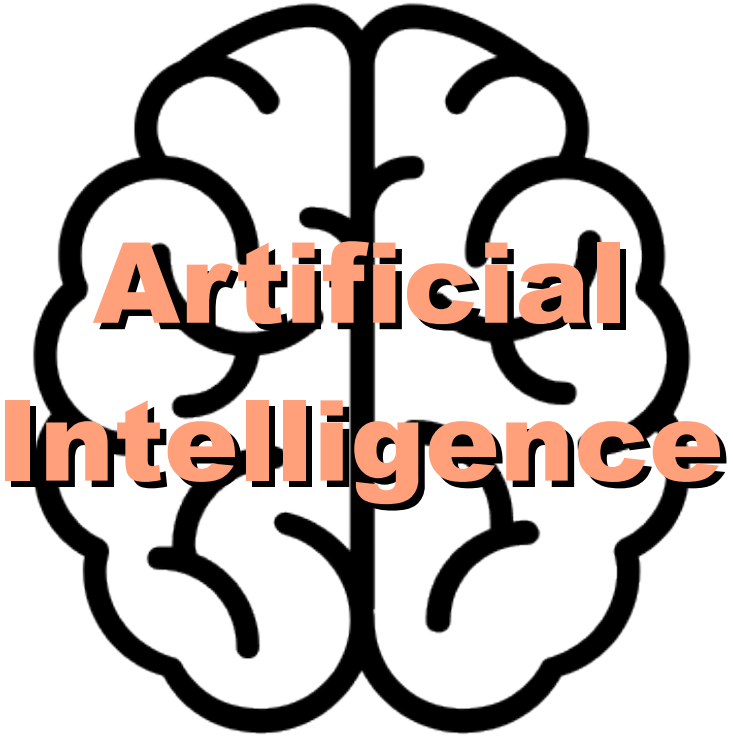
These Artifacts Aren't Fiction

Cloud Security

https://youtu.be/0vijJ7r-c8s

Cloud Security
and Other Assorted
Cautionary Tales

“The New Hotness”



Ways (A)I Benefitted

Daily Tasks

Communications
Note Taking
Summarizing
Research
Finding Things

IR Work

Code Analysis
Code/Query Generation
Log Analysis
Triage Guidance
Event Correlation

The Best Way to Learn

ON

Learning

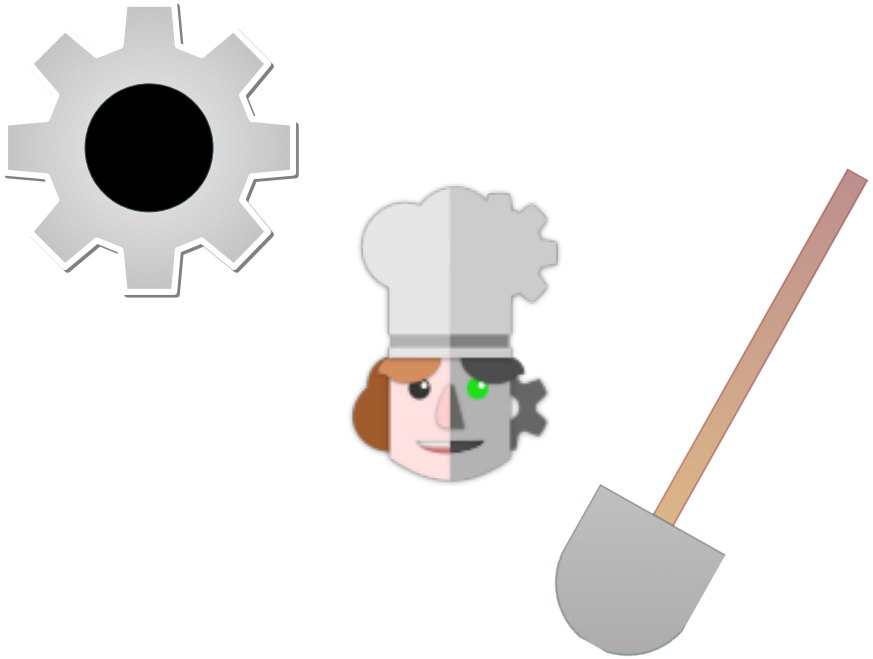
Network Packet Analysis

MATT SCHEURER

and the WIRESHARK WORKSHOP

http://workshops.dfirmatt.com

CyberChef



http://workshops.dfirmatt.com

Kusto Query Language

KQL

http://workshops.dfirmatt.com

Tools to Learn

- Incident Response & Forensics tools in
  - Kali Linux (Yes, really!)
  - REMnux (Linux)
  - SANS SIFT Workstation (Linux)
  - CSI Linux
  - FireEye's FLARE VM (Windows)

IR Ride Along

https://youtu.be/kjf-fBrSzVA

Questions

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```
if (badEvent == "true") {
  perform = "Incident Response";
}
```